English

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ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2024 - 2025)

Test Pattern

NEET(UG)
MAJOR
21-03-2025

PRE-MEDICAL : LEADER & ACHIEVER COURSE PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAAZ, MAAW, SPARK PRO

Test Booklet Code

L3

This Booklet contains **24** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case**.
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : ______________________

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YOUR TARGET IS TO SECURE GOOD RANK IN PRE-MEDICAL 2025

SUBJECT : BIOLOGY

Topic : SYLLABUS-3

1. **Statement-I :-** Photosynthesis is a physico-chemical process.
Statement-II :- Ultimately all living forms on earth depend on sunlight for energy.

(1) Only Statement I is correct and Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I & II are correct.
(4) Both Statement I & II are incorrect.

2. __A__ in 1770 performed a series of experiments that revealed the essential role of air in the growth of green plants.
Identify the A.

(1) Jan Ingenhousz
(2) T.W. Engelmann
(3) Julius Von sachs
(4) Joseph Priestley

3. Identify the correct statements :
A. T.W. Engelmann used prism and green algae, *Cladophora* placed in a suspension of aerobic bacteria.
B. All living forms on earth not depend on sun light for energy.
C. Variegated leaf experiment proves that green parts of plant and sunlight is essential for photosynthesis.
D. Julius von Sachs provided evidence for production of glucose when plants grow. Glucose is usually stored as starch.
Choose the correct answer from the options given below.

(1) A and D only
(2) A, C and D only
(3) B and C only
(4) A, B and C only

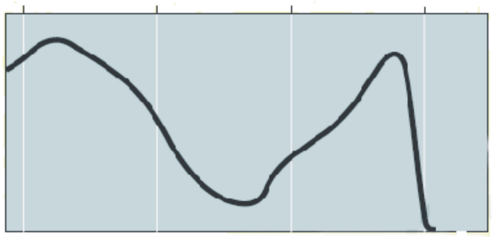
4. Where exactly enzymes of dark reaction are present within chloroplast ?

(1) Thylakoid membrane
(2) Grana
(3) Stromal lamella
(4) Stroma of chloroplast

5. Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) :- O_2 evolved by the green plant comes from H_2O not from CO_2.
Reason (R) :- In green plants H_2O is the hydrogen donor.
In the light of above statements, choose the most appropriate answer from the options given below :

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

6. Identify the below graph :



Rate of photosynthesis (measured by O_2 release)

(1) Action spectrum of photosynthesis
(2) Absorption spectrum of chlorophyll a
(3) Absorption spectrum of chlorophyll b
(4) Action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll a

7. **Assertion :-** There is no complete one to one overlapping between absorption spectrum of chlorophyll 'a' and the action spectrum of photosynthesis.
Reason :- Accessory pigments absorb wider range of wavelength and transfer energy to chlorophyll 'a'.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

8. Match the list-I with list-II.

List-I		List-II	
(a)	Light reaction	(i)	Yellow green
(b)	Chlorophyll b	(ii)	ATP + NADPH
(c)	Accessory Pigments	(iii)	Blue and Red region of spectrum
(d)	Maximum photosynthesis	(iv)	Chl b, xanthophyll and carotene

Choose the correct answer from the options given below :

(1) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)
(2) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)
(3) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)
(4) (a) – (i), (b) – (iv), (c) – (ii), (d) – (iii)

9. Select the correct statements :
(a) Pigments are substances that have an ability to absorb light at specific wavelength.
(b) Dark reaction occur in darkness only.
(c) We can separate the leaf pigments of any green plant through paper chromatography.
(d) Accessory pigments enable a wider range of wavelength of incoming light to be utilized for photosynthesis.

(1) a, b and c (2) a, c and d
(3) a and d only (4) c and d only

10. Match List I with List II and select the correct option.

List- I		List- II	
A.	NADP reductase enzyme	(i)	Transmembrane channel
B.	CF_0	(ii)	Inner side of the membrane
C.	CF_1	(iii)	Stroma side of membrane
D.	Splitting of the water molecule	(iv)	Protrudes on the outer surface of the thylakoid membrane

(1) A – ii, B – iv, C – iii, D – i
(2) A – iii, B – i, C – iv, D – ii
(3) A – iv, B – iii, C – ii, D – i
(4) A – iii, B – ii, C – iv, D – i

11. Match the column I with column II

	Column I		Column II
i	3-phosphoglyceric acid	a	First stable product in C_3 cycle
ii	Oxalo-acetic acid	b	First stable product in C_4 cycle
iii	RuBP	c	Primary CO_2 acceptor in C_3 cycle
iv	Phosphoenol pyruvate	d	Primary CO_2 acceptor in C_4 cycle

(1) a-i, b-ii, c-iii, d-iv
(2) a-ii, b-i, c-iii, d-iv
(3) a-iii, b-i, c-ii, d-iv
(4) a-iv, b-iii, c-ii, d-i

12. Which of the following characteristic is not shown by C_4 plants ?

(1) Kranz anatomy

(2) Can tolerate higher temperature

(3) Response to high light intensities

(4) Photorespiration.

13. Which of the following condition is favourable for photorespiration ?

(1) High light intensity (2) Low O_2

(3) High CO_2 (4) Low temperature

14. Given below are two statements :-

Statements–I :- For every CO_2 molecule entering the Calvin cycle, 3 molecules of ATP and 2 of NADPH are required.

Statements–II :- In C_4 plants photorespiration does not occur. This is because they have a mechanism that increases the concentration of CO_2 at the enzyme (RuBisCO) site.

In the light of the above statements, choose the correct answer form the options given below :

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

15. Major limiting factor for photosynthesis in terrestrial environment ?

(1) O_2 concentration (2) CO_2 concentration

(3) H_2O (4) Temperature

16. The plant or internal factors of photosynthesis depends on -

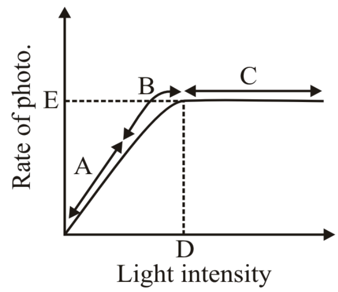
(1) Growth of plants

(2) Genetic predisposition

(3) Both (1) & (2)

(4) None of these

17. Recognise the figure & find out the correct match :-



	Column-I		Column-II
(i)	B-represents	(a)	Light is no longer limiting
(ii)	C-represents	(b)	Some factor other than light intensity is becoming the limiting factor
(iii)	D-represents	(c)	Maximum rate of photosynthesis
(iv)	E-represents	(d)	Light saturation point

(1) i-a, ii-b, iii-c, iv-d (2) i-c, ii-d, iii-b, iv-a

(3) i-b, ii-a, iii-d, iv-c (4) i-a, ii-b, iii-d, iv-c

18. Which one of the following statement is incorrect about cellular respiration in plants ?

(1) Breaking of complex compound through oxidation with in cell, lead to release of considerable amount of energy

(2) During respiration trapped energy is used for synthesis of ATP

(3) During respiration all energy contained in compounds is released in a single step.

(4) During respiration energy is released in a series of slow step wise reactions.

19. Which of the following is not a reason why plant can get along without respiratory organs ?

(1) Each plant part takes care of its own gas-exchange needs.

(2) Plants do not present great demands for gas exchange.

(3) Distance that gases must diffuse even in large, bulky plants is not great.

(4) Lenticels are present at each part of the plant.

20. The process that leads to a complete oxidation of organic substances in the presence of oxygen is :-

(1) Aerobic respiration

(2) Anaerobic respiration

(3) Both (1) and (2)

(4) Fermentation

21. Match the column :-

	Column-I		**Column-II**
(a)	Final product of glycolysis	(i)	In cytoplasm and mitochondria
(b)	Aerobic respiration	(ii)	In Yeast cell
(c)	Anaeobic respiration	(iii)	Pyruvate

(1) a-(iii), b-(ii), c-(i)

(2) a-(iii), b-(i), c-(ii)

(3) a-(ii), b-(i), c-(iii)

(4) a-(ii), b-(iii), c-(i)

22. The process of respiration common to both facultative anaerobe as well as obligate anaerobe is :-

(1) Glycolysis

(2) Link reaction

(3) Kreb's cycle

(4) Electron transport system

23. What will be net gain of ATP through substrate level phosphorylation during Glycolysis.

(1) 4 ATP

(2) 2 ATP

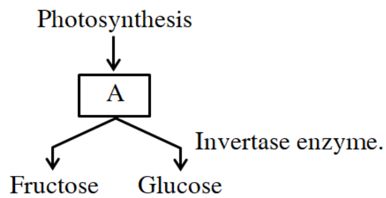
(3) 8 ATP

(4) 6 ATP

24. In glycolysis .....A..... undergoes .....B..... oxidation to form two molecules of .....C....... Find out A, B and C :-

	A	B	C
(1)	Glucose	Complete	Pyruvic acid
(2)	Glucose	Partial	PEP
(3)	Starch	Partial	Pyruvic acid
(4)	Glucose	Partial	Pyruvic acid

25. Study the flow chart and identify A



(1) Galactose

(2) Sucrose

(3) Maltose

(4) Lactose

26. Site of link reaction is –

(1) Cytoplasm

(2) Matrix of mitochondria

(3) Inner mitochondrial membrane

(4) Lumen of thylakoid

27. How many of ATP produced through one turn of Kreb's cycle during oxidation of glucose ?

(1) 30 ATP (2) 24 ATP

(3) 12 ATP (4) 38 ATP

28. Electron transport system and ATP synthase are present in which of the following part of mitochondria ?

(1) Outer membrane

(2) Inner membrane

(3) Matrix

(4) Peri mitochondrial space

29. The movement of electrons from NADH + H^+ occurs via complex :-

(a) Complex II - Succinate dehydrogenase

(b) Complex I - NADH dehydrogenase

(c) Complex III - Cytochrome bc_1

(d) Complex IV - Cytochrome C oxidase

(1) a ⟶ c ⟶ d

(2) b ⟶ c ⟶ d

(3) a ⟶ b ⟶ c ⟶ d

(4) a ⟶ d ⟶ c

30. At how many steps CO_2 is/are released in aerobic respiration ?

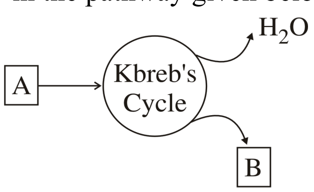
(1) One or two (2) Three

(3) Five (4) Twelve

31. How many ATP are produced through ETS during oxidation of one molecule of glucose ?

(1) 8 (2) 24 (3) 6 (4) 34

32. Find A & B in the pathway given below :-



(1) A - Pyruvic acid, B - O_2

(2) A - Acetyl CoA, B - O_2

(3) A - Acetyl CoA, B - CO_2

(4) A - Glycerol, B - O_2

33. Which of the following statement is incorrect ?

(1) R.Q for carbohydrates is one

(2) R.Q for fat is less than one

(3) R.Q for organic acids is more than one

(4) R.Q for protein is more than one

34. How many ATP are produced as net gain during anaerobic respiration when pyruvate is converted into lactic acid ?

(1) One (2) Two

(3) Three (4) Zero

35. Select the correct statement/s :-

(a) Growth can be defined as an irreverssible permanent increase in size of an organ or its parts.

(b) Growth is accompanied by metabolic process.

(c) The swelling of piece of wood when placed in water involves imbibition.

(1) Only a and b (2) Only b and c

(3) Only a and c (4) a, b & c

36. Which statements is/are correct in given below ?

(I) Plant retain the capacity for unlimited growth throughout their life

(II) Vascular cambium and cork cambium appear later in life in all plants.

(III) Vascular cambium and cork cambium are permanent tissues.

(IV) Interfascicular cambium and cork cambium are not capable of dividing.

(1) Only I (2) II and III

(3) II, III, IV (4) I and II

37. Cells in watermelon may increase in size by upto how many times?

(1) 17,500 times (2) 3,50,000 times

(3) 2,10,000 times (4) 200 times

38. Root elongation at constant rate examplify :

(1) Arithmetic growth (2) Geometrical growth

(3) Sigmoid growth (4) Both (2) and (3)

39. Which of the following is an intrinsic factor that affects plant growth ?

(1) Light

(2) Plant growth regulators

(3) Water

(4) Temperature

40. Which statement/statements are correct for plant growth ?

(A) At cellular level growth is increase in amount of the protoplasm.

(B) Cells in watermelon may increase in size by upto 17500 times.

(C) In a dorsiventral leaf, growth is measured by increase in surface area.

Select the correct option :-

(1) A & B (2) B & C

(3) A & C (4) A, B & C

41. Decapitation is generally done in tea plantations, and hedge making, what could be the possible reason for this ?

(1) Auxin is generally produced by growing apices which leads to apical dominance.

(2) More lateral branches required in tea plantations and hedge making.

(3) Decapitation helps the plants in growing taller.

(4) Both 1 and 2

42. Spraying juvenile conifers with GA_S hasten the maturity period thus leading to early production of which part of the plant ?

(1) Fruit (2) Root

(3) Seed (4) Stem

43. ABA plays a major role in :-

(1) seed development

(2) seed maturation

(3) seed dormancy

(4) All of the above

44. Photochemical phase of photosynthesis includes :

(1) light absorption (2) water splitting

(3) oxygen release (4) All of the above

45. **Statement – I :** The bakanae disease of rice seedlings, was caused by a fungal pathogen *Gibberella Fujikurai.*

Statement – II : The appearance of symptoms of the bakane disease is due to active substance cytokinin.

(1) Statement I and II are correct

(2) Statement I and II are incorrect

(3) Statement I is correct and II is incorrect

(4) Statement II is correct and I is incorrect

46. **Statement-I :** The afferent nerve fibres transmit impulses from organs to the CNS.

Statement-II : Visceral nervous system is the part of the peripheral nervous system.

(1) Both statements are incorrect.

(2) Both statements are correct

(3) Only statement-I is incorrect

(4) Only statement-II is incorrect

47. **Statement-I :** Neurons can detect, receive and transmit different kind of stimuli.

Statement-II : In *Hydra* the neural system is very simple and composed of brain and ganglia.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement-I is correct

(4) Only statement-II is correct

48. Read the following statement :

(a) Axonal membrane is comparatively less permeable to K^+ ions in resting stage.

(b) Axonal membrane is highly permeable for Na^+ ions in resting stage.

(c) Ionic gradients across the resting membrane are maintained by the sodium-potassium pump.

(d) Neurons are excitable cells because their membranes are in a polarised state.

Choose the **correct** option -

(1) a and b (2) c and d

(3) b and c (4) a and d

49. Corpora quadrigemina are part of :-

(1) Hind brain (2) Fore brain

(3) Mid brain (4) Spinal cord

50. Nerve impulses are initiated in neurons when its membrane become permeable to :-

(1) K^+ ion (2) Ca^{2+} ion

(3) Na^+ ion (4) Cl^- ion

51. Brain stem is made up of –

(1) Mid brain + pons + medulla oblongata

(2) Mid brain + Fore brain

(3) Fore brain + Pons

(4) Hind brain only

52. Sodium-potassium pump transport.

(1) $3Na^+$ out and $2K^+$ in

(2) $2Na^+$ out and $3K^+$ in

(3) $3Na^+$ in and $2K^+$ out

(4) $2Na^+$ in and $3K^+$ out

53. Which part of the CNS is responsible for regulating basic vital life functions such as breathing and heart beat ?

(1) Cerebellum (2) Medulla oblongata

(3) Cerebrum (4) Thalamus

54. "Amygdala" is a deeply associated structure present in ?

(1) Cerebellar hemisphere

(2) Cerebral hemisphere

(3) Hypothalamus

(4) Medulla oblongata

55. Unmyelinated nerve fibre is :

(1) Not enclosed by a Schwann cell hence no myelin sheath

(2) Enclosed by a Schwann cell that does not form a myelin sheath

(3) Not enclosed by a Schwann cell but enclosed by a myelin sheath formed by Oligodendrocyes.

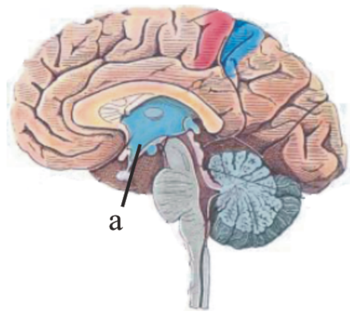
(4) Only found in the central neural system.

56. Memory & communications is the function of :-

(1) Hypothalamus (2) Association area

(3) Thalamus (4) Amygdala

57. Which one is not related with 'a' ?



(1) Regulation of sexual behaviour

(2) Neurosecretory cells

(3) Intersensory associations and balance

(4) Regulation of body temperature

58. Cerebral cortex are greyish in appearance. It is mainly due to :

(1) Bundle of Axon

(2) Corpus callosum

(3) Concentration of neuron cell bodies

(4) Presence of myelin sheath

59. How many of the following function are related with limbic system ?

(a) Motivation
(b) Rage
(c) Pleasure
(d) Excitement

(1) 3 (2) 4 (3) 2 (4) 1

60. Glucocorticoids stimulates :-

(A) Gluconeogenesis (B) Lipolysis
(C) Proteolysis

(1) Only A and B
(2) Only A
(3) Only B
(4) All A, B and C

61. Hormone responsible for development of alveolar parts of mammary gland :

(1) Estrogen
(2) Progesterone
(3) Prolactin
(4) Oxytocin

62. Which one of the following hormones in general is not involved in sugar metabolism ?

(1) Glucagon
(2) Cortisol
(3) Aldosterone
(4) Insulin

63. Which one of the following pair of hormones, are those that can easily pass through the cell membrane of the target cell ?

(1) Insulin, glucagon
(2) Thyroxine, insulin
(3) Somatostatin, oxytocin
(4) Cortisol & Testosterone

64. Identify A, B and C from table :-

Thymosin	Thymus gland	(A)
(B)	Pancreas	Secretion inhibited by somatostatin
Cortisol	(C)	Suppress immunity during organ transplantation

	C	B	A
(1)	Adrenal cortex	Thyroxin	Antibody formation by T-lymphocyte
(2)	Testis	Somatostatin	Ca^{+2} level maintain in blood
(3)	Adrenal medulla	Glucagon	Maturation of B lymphocyte
(4)	Adrenal cortex	Insulin	Maturation of T-lymphocyte

65. The hormone responsible for regulation of calcium and phosphorous metabolism is secreted by :–

(1) Pancreas (2) Pineal
(3) Thymus (4) Parathyroid

66. Which hormone is not produced in Intestine

(1) CCK (2) Gastrin
(3) GIP (4) Secretin

67. **Assertion :** F.S.H and L.H are released from anterior pituitary due to secretions from hypothalamus.

Reason : GnRH is released from hypothalamus and works on anterior pituitary.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

68. Which one is not a secondary messenger ?

(1) cAMP (2) IP_3 (3) Ca^{+2} (4) K^+

69. **Statement-I :-** Aldosterone helps in the maintenance of blood pressure

Statement-II :- Aldosterone acts mainly at renal tubule and stimulate reabsorption of K^+ and water.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

70. Match List-I with List-II.

	List-I		List-II
(a)	Aldosterone	(i)	RBC production
(b)	Androgenic steroid	(ii)	Osmotic pressure
(c)	Cortisol	(iii)	Pupillary dilation
(d)	Catecholamines	(iv)	Growth of pubic hair

Choose the most appropriate answer from the options given below :-

(1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

71. Acromegaly is a disease caused by :

(1) Over secretion of growth hormone in childhood

(2) Over secretion of growth hormone in adulthood

(3) Under secretion of growth hormone in adulthood

(4) Deficiency of calcium and phosphorous in the diet.

72. Which one of the following hormones though synthesised elsewhere, is stored and released by the pituitary gland ?

(1) Melanocyte stimulating hormone

(2) Antidiuretic hormone

(3) Luteinizing hormone

(4) Prolactin

73. Parathormone (parathyroid hormone) act on bone and stimulate which of the following action ?

(1) Carbohydrate metabolism

(2) Bone resorption

(3) Bone formation

(4) Fat metabolism

74. During muscles contraction ATP bind with :-

(1) Actin (2) Troponin

(3) Myosin (4) Tropomyosin

75. Cardiac muscles are :-

(1) Involuntary, Unbranched, Striated

(2) Voluntary, Unbranched, Nonstriated

(3) Involuntary, Branched, Striated

(4) Voluntary, Branched, Nonstriated

76. Match List-I with List-II :-

	List-I		List-II
(A)	Cranium	(I)	Gliding joint
(B)	Between adjacent vertebrae	(II)	Pivot joint
(C)	Between carpals	(III)	Fibrous joint
(D)	Between atlas and axis	(IV)	Cartilagenous joint

(1) A-I, B-II, C-III, D-IV (2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-IV, D-II (4) A-II, B-IV, C-I, D-III

77. Which of following is a wrong statement, regarding muscle contraction ?

(1) I-band gets reduced

(2) A-band retain the same length

(3) H-zone remains unchanged

(4) Sarcomere shortens

78. Incorrect about skeleton muscles is :-

(1) They are syncytium.

(2) They have light and dark bands.

(3) Based on appearance skeleton muscles are unstriated

(4) They are voluntary in nature

79. Which joint is similar to elbow joint ?

(1) Joint between humerus and scapula

(2) Joint between atlas and axis

(3) Joint between Femur, Tibia and patella

(4) Joint between Radius and Ulna

80. Joint between carpal and metacarpals of thumb :

(1) Saddle

(2) Hinge

(3) Pivot

(4) Ball and socket

81. Pelvic girdle consists of :-

(1) 1 coxal bone

(2) 2 coxal bone

(3) 3 coxal bone

(4) 6 coxal bone

82. Acromion process present on ?

(1) Scapula (2) Clavicle

(3) Humerus (4) Femur

83. The example of fibrous joint is ?

(1) Vertebral column

(2) Cranium

(3) Knee

(4) Elbow

84. Which of the following is incorrect match.

(1) Ear ossicles- Malleus, Incus and stapes

(2) Floating ribs-8^{th}, 9^{th} and 10^{th}

(3) Pectoral girdle- Scapula and clavicle

(4) Pelvic girdle- Ilium, ischium and pubis

85. Which statement is not correct about osteoporosis ?

(1) In this, bone mass is decreased

(2) It is an age related disorder

(3) Increased levels of estrogen is a common cause of it.

(4) In this, chances of fractures are increased.

86. In voluntary muscle "Fascia" term is used for

(1) A collagenous connective tissue by which muscle bundles are held together

(2) A elastic protein which present between I-Bands

(3) A calcium rich structure in sarcoplasm

(4) Central gap between thick filaments.

87. Which structure of muscle is an active ATPase enzyme.

(1) Globular head of meromyosin

(2) Globular head of actin

(3) 'F' - actin helically wound to each other

(4) Complex troponin protein

88. Character of white muscle fibres is -

(1) anaerobic process for energy

(2) amount of Sarcoplamic reticulum is high

(3) Few mitochondria

(4) All of these

89. What is the main reason for tetany?

(1) High level of Ca^{++} ion in bones.

(2) Low level of Ca^{++} ion in bones.

(3) High level of Ca^{++} ion body fluids.

(4) Low level of Ca^{++} ion in body fluids.

90. Which of the following is a genetic disorder?

(1) Myasthenia gravis (2) Muscular dystrophy

(3) Tetany (4) Gout

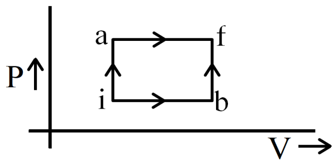
SUBJECT : PHYSICS

Topic : SYLLABUS-3

91. Oxygen and hydrogen both are at same temperature T. The ratio of mean kinetic energy of oxygen molecules to that of the hydrogen molecules will be :

(1) 16 : 1 (2) 1 : 1 (3) 4 : 1 (4) 1 : 4

92. When a system is taken from state 'i' to state 'f' along path 'iaf', it is found that Q = 50 cal and W = 20 cal. Along the path 'ibf' Q = 36 cal, then 'W' along this path (Q is heat supplied to gas and W is work done by gas) :-



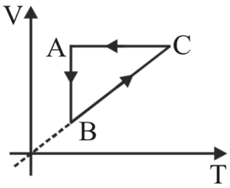
(1) 6 cal (2) 16 cal

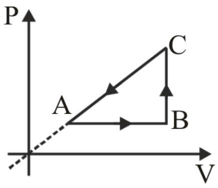
(3) 66 cal (4) 14 cal

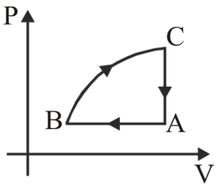
93. A carnot engine operates with source at 127 °C and sink at 27 °C. If source supplies 40 kJ of heat energy, the work done by engine is :-

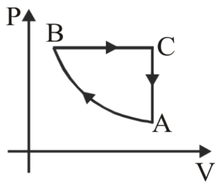
(1) 30 kJ (2) 10 kJ (3) 4 kJ (4) 1 kJ

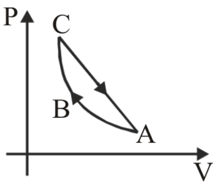
94. Cyclic process ABCA is shown in V–T diagram corresponding process on the P–V diagram is :



(1) 

(2) 

(3) 

(4) 

95. A gas is contained in a container at pressure 10 kPa. When gas is expanded isothermally from volume $9m^3$ to $27m^3$ work done will be.

(1) 27×10^3 (ℓn 3)

(2) 9×10^4 (ℓn 3)

(3) $9 \times 10^3 \left(\ell n \frac{1}{3}\right)$

(4) $27 \times 10^3 \left(\ell n \frac{1}{3}\right)$

96. Two spheres are made of same metal and have same mass. One is solid and other is hollow. When heated to the same temperature, which of the following statements is correct about the percentage increase in their diameter?

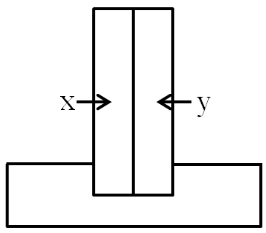
(1) It will be more for hollow sphere

(2) It will be more for solid sphere

(3) It will be same for both spheres

(4) It may be more or less for any sphere depending upon the ratio of the diameters of the two spheres

97. A bimetallic strip consists of metals x and y. It is mounted rigidly at the base as shown in figure. The metal x has a higher coefficient of expansion as compared to that for metal y. When the bimetallic strip is placed in cold bath?



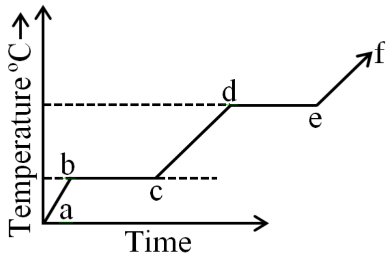
(1) It will bend toward the right

(2) It will bend toward the left

(3) It will not bend but shrink

(4) It will neither bend nor shrink

98. The following figure represents the temperature versus time plot for a given amount of a substance when heat energy is supplied to it at a fixed rate and at a constant pressure. Which part of the below plot represents a phase change?



(1) a to b and e to f (2) b to c and c to d

(3) d to e and e to f (4) b to c and d to e

99. A body cools from 60°C to 50°C in 10 minutes. If the room temperature is 25°C and assuming Newton's Law of cooling to hold good, the temperature of the body at the end of the next 10 minutes will be :

(1) 38.5°C (2) 40°C

(3) 42.85°C (4) 45°C

100. Gravitation is necessary for -

(1) For conduction

(2) For natural convection

(3) For radiation

(4) None of the above

101. Total radiation energy incident on body is 400 Joule. If 30% of incident radiation reflected back and 120 J is absorbed by body. Then find out transmittive power in percentage.

(1) 60% (2) 40%

(3) 50% (4) None

102. For a black body λ_{max} is 300 Å at 727°C. Find λ_{max} for same black body at 1227°C.

(1) 500 Å (2) 1000 Å

(3) 1500 Å (4) 200 Å

103. A black body at a temperature of 227°C radiates heat at the rate 7 cal/cm^{2}sec. At temperature of 727°C, the rate of heat radiated in the same units will be :-

(1) 50 (2) 112 (3) 80 (4) 60

104. Under steady state the temperature of a body :

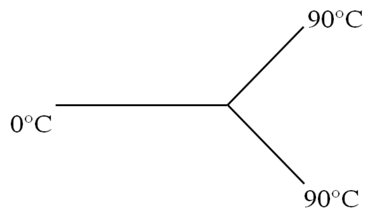
(1) Increases with time

(2) Decreases with time

(3) Doesn't change with time and is same at all points of the body

(4) Doesn't change with time but is different at different cross - sections of the body.

105. Three rods made of the same material and having the same cross - section have been joined as shown in figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C respectively. The temperature of the junction of the three rods will be :



(1) 45°C (2) 60°C

(3) 30°C (4) 20°C

106. A mass m executes simple harmonic motion with frequency n when suspended from a weightless spring. The spring is cut into two equal parts and an another mass 2m is suspended from one of the part. The frequency of oscillation of this mass 2 m is :-

(1) n (2) 2n

(3) $\frac{n}{\sqrt{2}}$ (4) n(2)^2

107. Infinite springs with force constant k, 2k, 4k, 8k ..... respectively are connected in series combination. The effective force constant will be :-

(1) 2k (2) k (3) k/2 (4) 4k

108. Two simple harmonic motions represented by $y_1 = 4\sin(4\pi t + \frac{\pi}{2})$ and $y_2 = 3\cos(4\pi t)$ superimpose over each other then the resultant amplitude is (all quantities are in SI units) :-

(1) 7 m (2) 1 m

(3) $(2+\sqrt{3})$ m (4) $(2-\sqrt{3})$ m

109. The angle made by the string of a simple pendulum with vertical depends upon time as $\theta = \frac{\pi}{90}\sin\pi t$. Find the length (in metre) of the pendulum. (If $g = \pi^2 m/s^2$).

(1) 1 (2) 2 (3) 3 (4) 4

110. For S.H.M. of a particle, amplitude is 6cm. At a certain instant its potential energy is half of the total energy then the distance of particle at this instant from its mean position is :-

(1) 3 cm

(2) 4.2 cm

(3) 5.8 cm

(4) 6 cm

111. A simple pendulum whose time period is 2 sec at the surface of earth is now suspended in a satellite orbiting around the earth at a height 3R, where R is the radius of earth. The time period of this pendulum is :-

(1) zero

(2) $2\sqrt{3}$ sec

(3) 4 sec

(4) infinite

112. The time period of a simple pendulum is 2 sec. If its length becomes 16 times, then its time period becomes :-

(1) 16 sec (2) 12 sec

(3) 8 sec (4) 4 sec

113. The kinetic energy of a particle executing S.H.M is 16 J, when it is in its mean position. If the amplitude of oscillation is 25 cm and the mass of the particle is 5.12 kg, the time period of its oscillation is :-

(1) $\frac{\pi}{5}$ sec (2) 2π sec.

(3) 20π sec (4) 5π sec

114. The particle is executing SHM on a line 4 cm long. If its velocity at mean position is 12cm/s, its frequency (in hertz) will be -

(1) $\frac{2\pi}{3}$ (2) $\frac{3}{2\pi}$ (3) $\frac{\pi}{3}$ (4) $\frac{3}{\pi}$

115. Which of the following relationship between acceleration 'a' and displacement 'x' of a particle represent the simple harmonic motion :-

(1) $a = 2x$ (2) $a = -25x$

(3) $a = 4x^2$ (4) $a = -100x^2$

116. The displacement of a particle performing linear simple harmonic motion of amplitude A in one time period is :-

(1) A (2) 2A

(3) zero (4) 3A

117. For S.H.M match the following

	Column-I		Column-II
A	Displacement and velocity	P	Phase difference is zero
B	Displacement and acceleration	Q	Phase difference is $\pi/2$
C	Velocity and acceleration	R	Phase difference is π

(1) $A \rightarrow P, B \rightarrow Q, C \rightarrow R$

(2) $A \rightarrow Q, B \rightarrow R, C \rightarrow Q$

(3) $A \rightarrow Q, B \rightarrow P, C \rightarrow R$

(4) $A \rightarrow R, B \rightarrow P, C \rightarrow R$

118. **Assertion** :- If a pendulum is suspended in a lift and lift accelerates upwards, then its time period will decrease as compared to that when it is in stationary lift.

Reason :- Effective value of acceleration due to gravity will be $g_{eff} = g + a$ in this case

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

119. The equation of a stationary wave is

$y = 0.8 \cos\left(\frac{\pi x}{20}\right) \sin 200\pi t$, where x and y are in cm and t is in second. The separation between consecutive nodes will be :-

(1) 20 cm (2) 10 cm

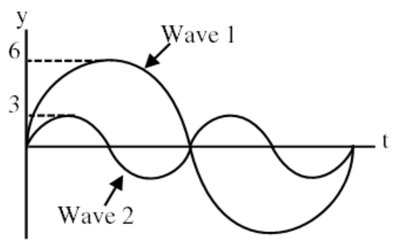
(3) 40 cm (4) 30 cm

120. The fundamental frequency of a closed organ pipe is 220 Hz. If $\frac{1}{4}$ of the pipe is filled with water, the frequency of the first overtone of the pipe now is :-

(1) 220 Hz (2) 440 Hz

(3) 880 Hz (4) 1760 Hz

121. Two waves propagating in same medium are represented by y-t curves in the figure. The ratio of their average intensities is :-



(1) 4 (2) 1/4

(3) 9 (4) 1

122. When the string of a sonometer of length L between the bridges vibrates in the second overtone, the amplitude of vibration is maximum at :-

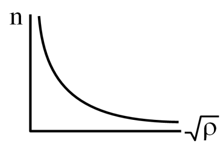
(1) $\frac{L}{2}$

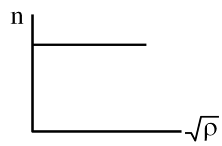
(2) $\frac{L}{4}$ and $\frac{3L}{4}$

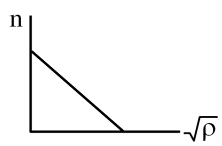
(3) $\frac{L}{6}$ and $\frac{3L}{6}$

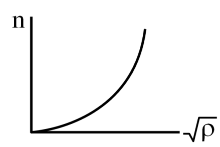
(4) $\frac{L}{8}$, $\frac{3L}{8}$ and $\frac{5L}{8}$

123. The correct graph between the frequency (n) of wave in a stretched wire and square root of density (ρ) of a wire, keeping its length, radius and tension constant, is :-

(1) 

(2) 

(3) 

(4) 

124. What is the phase difference, at a given instant of time, between two particles 25m apart, when the wave $y = 0.03 \sin \pi(2t - 0.01x)$ travels in a medium (x, y is in m and t in second) ?

(1) $\frac{\pi}{8}$ (2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$ (4) π

125. A string of linear mass density 4g/cm is vibrating according to equation

$y = A \sin(120\pi t) \cos\left(\frac{2\pi}{5}x\right)$, where x and y are in cm. The tension in the string is :-

(1) 3.6 N

(2) 36 N

(3) 7.2 N

(4) 72 N

126. A tuning fork T_1 produces 4 beats with tuning fork T_2 of frequency 256 Hz. When T_1 is filed, beats are found to occur at shorter intervals. What was the original frequency of tuning fork T_1 ?

(1) 260 Hz (2) 252 Hz

(3) 254 Hz (4) 258 Hz

127. An astronaut cannot hear his companion at the surface of the moon because :-

(1) Produced frequencies are above the audible frequencies

(2) There is no medium for sound propagation

(3) Temperature is too low during night and high during day

(4) There are too many craters on the surface of the moon

128. Which of the following properties of a wave does not change with change in medium ?

(1) Frequency (2) Wavelength

(3) Velocity (4) Amplitude

129. Two waves of intensities I_1 and I_2 pass through a region at the same time in the same direction superimpose on each other, the sum of the maximum and minimum intensities is :-

(1) $\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$ (2) $2(I_1+I_2)$

(3) I_1+I_2 (4) $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

130. A wave is represented by the relation

$$y = a_0 \sin 2\pi\,(nt - x/\lambda)$$

if the maximum particle velocity is three times the wave velocity, the wavelength λ of the wave is :-

(1) $\frac{\pi a_0}{3}$ (2) $\frac{2\pi a_0}{3}$

(3) πa_0 (4) $\frac{\pi a_0}{2}$

131. The equation of a travelling wave is

$y = 60\cos(1800t - 6x)$

where x and y is in metre and t is in sec. The ratio of maximum particle velocity to velocity of wave propagation is :-

(1) 3.6 (2) 36 (3) 360 (4) 0.36

132. 25 tuning forks are arranged in series in the order of decreasing frequency. Any two successive forks produce 3 beats/sec. If the frequency of the first tuning fork is the octave of the last fork, then the frequency of the 21^{st} fork is :-

(1) 72 Hz (2) 78 Hz

(3) 84 Hz (4) 87 Hz

133. An open organ pipe 40 cm long and a closed organ pipe 31 cm long, both having same diameter, are producing their first overtone, and these are in unison. The end correction of closed organ pipe is :-

(1) 1 cm (2) 2 cm

(3) 1.5 cm (4) 3 cm

134. Which of the following changes with time when damping is considered for a S.H.M. : -

(A) Time period (B) Angular frequency

(C) Amplitude (D) Energy

(1) A & B

(2) C & D

(3) B & C

(4) A, B, C & D

135. If in a resonance tube a oil of density higher than that of water is used then the resonance frequency would be :-

(1) Increased

(2) Decreased

(3) Slightly increased

(4) Remain the same

SUBJECT : CHEMISTRY

Topic : SYLLABUS-3

136. Number of π bonds in ethylene chloride.

(1) 0 (2) 1 (3) 2 (4) 3

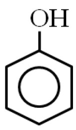
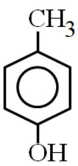
137. The number of ether isomers represented by formula $C_4H_{10}O$ is (only structural) :

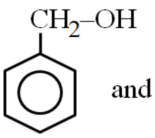
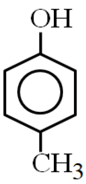
(1) 4 (2) 3 (3) 2 (4) 7

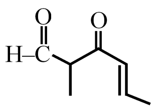
138. Which of the following pair are isomers ?

(1) CH_3-CH_2-OH and $CH_3-\underset{}{\overset{OH}{\overset{|}{C}}}H-CH_3$

(2) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ and $CH_3-CH_2-O-CH_3$

(3)  and 

(4)  and 

139. The IUPAC name of the compound  is.

(1) 5-formylhex-2-en-3-one

(2) 5-methyl-4-oxohex-2-en-5-al

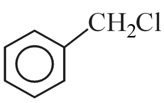
(3) 3-keto-2-methylhex-5-enal

(4) 3-keto-2-methylhex-4-enal

140. Which of the following IUPAC name is correct:

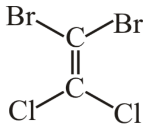
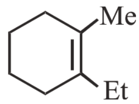
(1) CCl_4 carbon terachloride

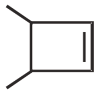
(2) CH_2Cl_2 methylene chloride

(3)  chlorophenylmethane

(4) $CH_2 = CH - CH_2 - Br$ Allyl bromide

141. Which of the following molecule can show geometrical isomerism ?

(1)  (2) 

(3)  (4) 

142. **Assertion :** Staggered conformation of ethane is most stable while eclipsed conformation is least stable.

Reason : Staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain

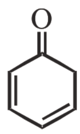
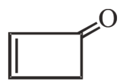
(1) Assertion is correct, reason is correct, reason is a correct explanation for assertion

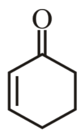
(2) Assertion is correct, reason is correct, reasion is not a correct explanation for assertion

(3) Assertion is correct, reason is incorrect.

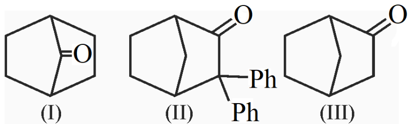
(4) Assertion is incorrect, reason is correct.

143. Which of the following compound will have most stable enol ?

(1)  (2) 

(3)  (4) 

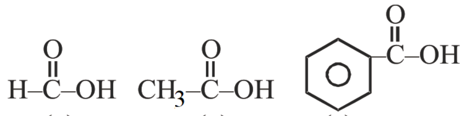
144. In which of the following compound tautomerism takes place :



(1) Both (I) and (II) (2) Both (II) and (III)

(3) (III) Only (4) Both (I) and (III)

145. Arrange the following acids in decreasing order of acidic strength :-



(x) (y) (z)

(1) x > z > y (2) x > y > z

(3) z > x > y (4) z > y > x

146. Correct order of –I effect is :-

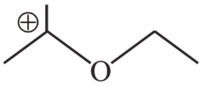
(1) –CN > –COOH > $-NO_2$

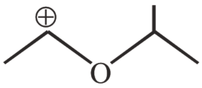
(2) –COOH > –CN > $-NO_2$

(3) $-NO_2$ > –COOH > –CN

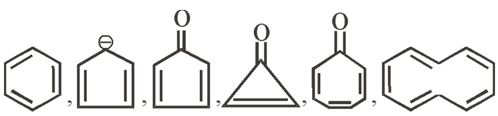
(4) $-NO_2$ > –CN > –COOH

147. Most stable carbocation is :-

(1)  (2) 

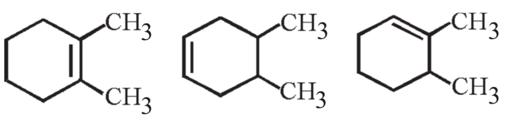
(3)  (4) 

148. Consider the following compounds / species.



The number of compounds which obey Huckel's rule is :-

(1) 6 (2) 4 (3) 5 (4) 3

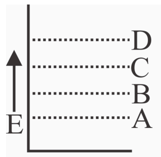
149. 

(I) (II) (III)

Which of following orders is correct for heat of hydrogenation of these compounds ?

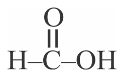
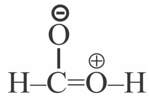
(1) I > II > III (2) III > II > I

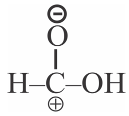
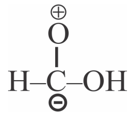
(3) II > III > I (4) III > I > II

150. 

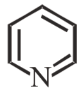
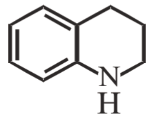
Formic acid

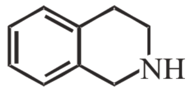
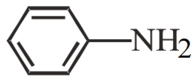
A, B, C, D are resonating structures of formic acid. Structure of B is:

(1)  (2) 

(3)  (4) 

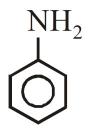
151. The compound that is easiest to protonate is ?

(1)  (2) 

(3)  (4) 

152. Correct decreasing order of basic strength :-

(A) $(C_2H_5)_2NH$ (B) $C_2H_5NH_2$

(C) $(C_2H_5)_3N$ (D) 

(1) A > B > C > D (2) D > A > C > B

(3) D > C > B > A (4) A > C > B > D

153. Given below are two statements

Statement-I : The energy of actual structure of the molecule (the resonance hybrid) is lower than that of any of the canonical structure.

Statement-II : The difference in energy between the actual structure and the highest energy resonance structure is called the resonance stabilisation energy.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I incorrect and statement-II is correct.

(4) Statement-I correct and statement-II is incorrect.

154. Match List - I with List - II.

	List-I (Compounds)		List-II (pK_a-Value)
(a)	o-Cresol	(i)	10
(b)	m-Cresol	(ii)	15.9
(c)	Ethanol	(iii)	10.1
(d)	Phenol	(iv)	10.2

(1) a-(iv), b-(iii), c-(ii), d(i)

(2) a-(iv), b-(ii), c-(iii), d(i)

(3) a-(iii), b-(iv), c-(ii), d(i)

(4) a-(ii), b-(iv), c-(i), d(iii)

155. A liquid compound (x) can be purified by steam distillation only if it is

(1) Steam volatile, immiscible with water

(2) Not steam volatile, miscible with water

(3) Steam volatile, miscible with water

(4) Not steam volatile, immiscible with water

156. Match List-I with List-II.

	List-I		List-II
(a)	Separation of aniline-water mixture	(i)	Fractional distillation
(b)	Separation of aniline-chloroform mixture	(ii)	Distillation under reduced pressure
(c)	Separation of Glycerol from spent-lye	(iii)	Simple distillation
(d)	Separation of different fractions of crude oil	(iv)	Steam distillation

Choose the correct answer from the options given below :-

(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

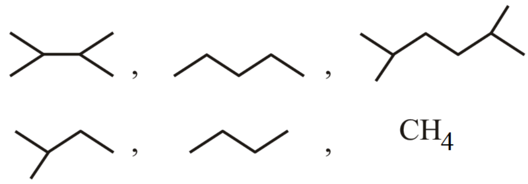
(4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

157. 0.1 gm of organic compound was analysed by Kjeldahl's method. In analysis produced NH_3 absorbed in 30 ml $\frac{M}{10}$ H_2SO_4. The remaining acid required 20 ml N/10 NaOH for neutralisation. Calculate percentage of nitrogen in organic compound :-

(1) 28 % (2) 56 %

(3) 35 % (4) 70 %

158. How many compound can be obtained in good yield by wurtz reaction.



CH_4

(1) 6

(2) 3

(3) 4

(4) 5

159. Match column I with column II

	Column-I (Reaction)		Column-II (Product)
I.	$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2O_3}$	P.	$2CH_3OH$
II.	$(CH_3)_3CH \xrightarrow{KMnO_4}$	Q.	$2CH_3COOH + 2H_2O$
III.	$2CH_4 + O_2 \xrightarrow{Cu/523K/100atm.}$	R.	$HCHO + H_2O$
IV.	$2CH_3CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3COO)_2Mn}$	S.	$(CH_3)_3C-OH$

(1) I–R, II–P, III–Q, IV–S

(2) I–P, II–S, III–R, IV–Q

(3) I–R, II–S, III–P, IV–Q

(4) I–R, II–P, III–S, IV–Q

160. Match the reactions in column-I with their types in column-II

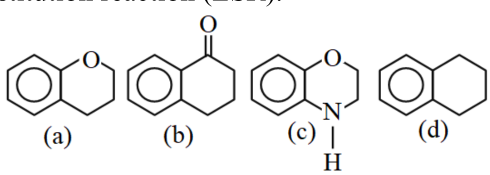
	Column-I (Reaction)		Column-II (Types)
(a)	$CH_3-CH_2-OH \xrightarrow[\Delta]{H^+}$	(p)	Electrophilic addition reaction
(b)	$CH_3-CH=CH_2 \xrightarrow[\text{Peroxide}]{HBr}$	(q)	Free radical addition reaction
(c)	 $\xrightarrow[\text{Dark}]{Cl_2/Fe}$	(r)	Elimination reaction
(d)	 $\xrightarrow{\text{Conc. } H_2SO_4}$	(s)	Electrophilic substitution reaction

(1) a-r, b-q, c-s, d-p (2) a-q, b-s, c-r, d-p
(3) a-p, b-r, c-s, d-q (4) a-s, b-p, c-q, d-r

161. Which of the following alcohol on dehydration with conc. H_2SO_4 at 440 K will give isobutylene.
(I) Isobutyl alcohol
(II) Tert. butyl alcohol
(III) n-butyl alcohol
(IV) Isopropyl alcohol
(1) I, III (2) II, III (3) III, IV (4) I, II

162. **Statement-I:** The presence of weaker π bonds make alkenes less stable than alkanes.
Statement-II: The strength of the double bond is greater than that of carbon-carbon single bond.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

163. Rank the following compounds in decreasing order of reactivity towards electrophilic aromatic substitution reaction (ESR).



(1) (c) > (a) > (b) > (d) (2) (c) > (a) > (d) > (b)
(3) (d) > (c) > (b) > (a) (4) (a) > (c) > (d) > (b)

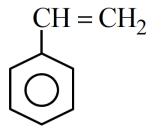
164. Primary alcohol can be formed as major product by :-

(1) $CH_3-\underset{CH_3}{\underset{|}{C}}=CH_2 \xrightarrow[\text{(ii) } H_2O_2/OH^{\ominus}]{\text{(i) } BH_3.THF}$

(2) $CH_3-\underset{CH_3}{\underset{|}{C}}=CH_2 \xrightarrow{\text{dil. } H_2SO_4}$

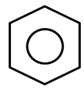
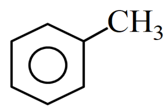
(3) $CH_3-\underset{CH_3}{\underset{|}{C}}=CH_2 \xrightarrow[\text{(ii) } NaBH_4]{\text{(i) } (CH_3COO)_2Hg.HOH}$

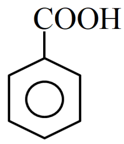
(4) All of these

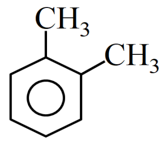
165. 

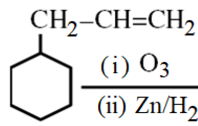
$\xrightarrow[H^+]{KMnO_4} A \xrightarrow[\Delta]{NaOH + CaO} B \xrightarrow[AlCl_3]{CH_3-Cl} C$

Identify C ?

(1)  (2) 

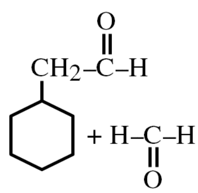
(3) 

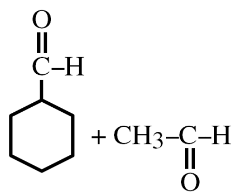
(4) 

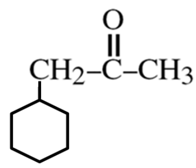
166. 

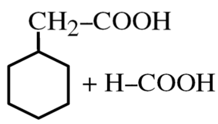
$\xrightarrow[\text{(ii) } Zn/H_2O]{\text{(i) } O_3}$ (Major product) :-

Major product will be :-

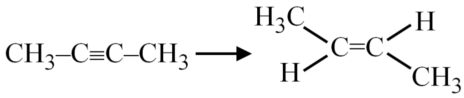
(1) 

(2) 

(3) 

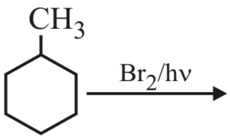
(4) 

167. The most suitable reagent for the following conversion is :-



(1) Na/liquid NH_3 (2) H_2, Pd/C, Quinoline

(3) Zn/HCl (4) Hg^{+2}/H^+, H_2O

168.  Monobromo derivatives.

The number of possible monobromo product is (excluding stereoisomers)

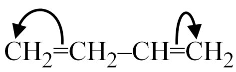
(1) 4 (2) 5 (3) 8 (4) 10

169. Which of the following alkyl bromides will eliminate HBr at fastest rate in elimination reaction.

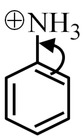
(1) Ethyl bromide (2) Propyl bromide

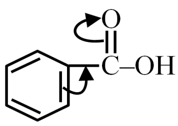
(3) Isopropyl bromide (4) t-Butyl bromide

170. Which of the following is correct representation of delocalisation of electrons during resonance :

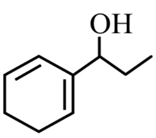
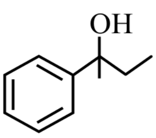
(1) 

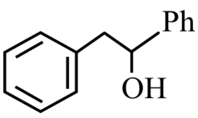
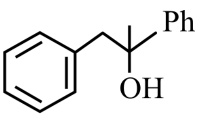
(2) 

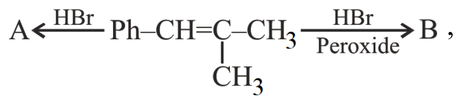
(3) 

(4) 

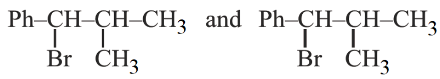
171. Which of the following alcohol is benzylic as well as secondary :-

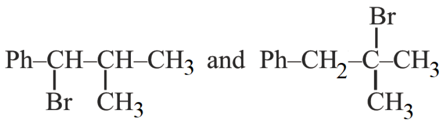
(1)  (2) 

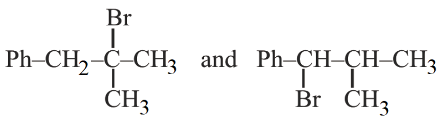
(3)  (4) 

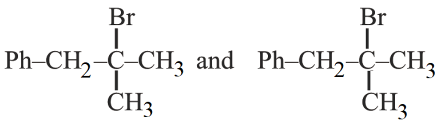
172. 

A and B is :-

(1) 

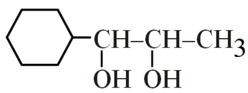
(2) 

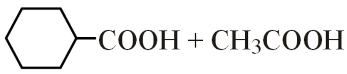
(3) 

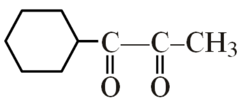
(4) 

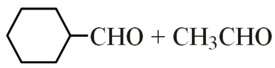
173. 

What are product in above reaction ?

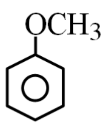
(1) 

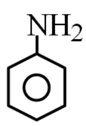
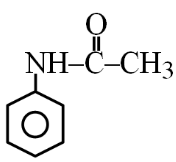
(2) 

(3) 

(4) 

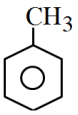
174. Which of the compound will not undergo Friedel craft's reaction -

(1)  (2) 

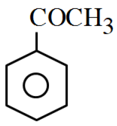
(3)  (4) 

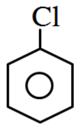
175. $n-\text{hexane} \xrightarrow[\substack{773K \\ 10-20\,atm}]{Cr_2O_3} A \xrightarrow[\text{Anhyd.}AlCl_3]{(CH_3CO)_2O} B$

Product B is

(1) 

(2) $CH_3-CH_2-CH_2-Cl$

(3) 

(4) 

176. Which of the following compound will not give a precipitate with Tollen's, reagent :-

(1) Ethyne (2) 1-Butyne

(3) 2-Pentyne (4) 1-Pentyne

177. Given below are two statements :-

Statement-I :- The boiling point of three isomeric pentanes follows the order

n-pentane > isopentane > neopentane

Statement-II :- When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the most appropriate answer from the option given below :

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

178.  $\xrightarrow[UV, 500K]{3Cl_2}$ 'A' (major product).

Number of σ bonds in 'A' is :-

(1) 6 (2) 12 (3) 18 (4) 24

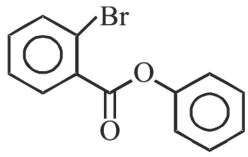
179. How many of following are ortho-para directing groups in electrophilic substitution reaction

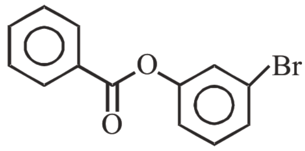
$-OH$, $-Cl$, $-NH-\overset{O}{\overset{\|}{C}}-CH_3$

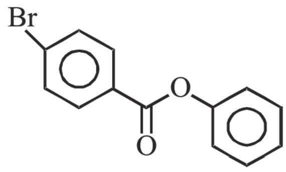
$-O-\underset{O}{\underset{\|}{C}}-CH_3$, $-\underset{O}{\underset{\|}{C}}-O-CH_3$, $-CH=O$

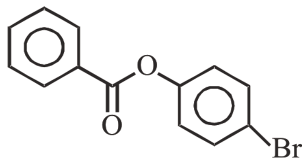
(1) 4 (2) 6 (3) 5 (4) 3

180. The major product formed on monobromination of phenylbenzoate is :-

(1) 

(2) 

(3) 

(4) 

SPACE FOR ROUGH WORK